

JavaScript: The New Parts

This article is the fourth part in this series. The author discusses the *for...of* iterator, collections and arrow functions.

The JavaScript language is evolving to enable the development of complex and large applications. This is a multi-year effort by ECMA (a standards body). In the previous article or third part of this series, we learnt about computed property keys and string functions. Their support is available in most recent versions of browsers. In this part, we will learn about a control structure for collections, new collection data structures and abbreviated syntax for function-definition using the arrow symbol ($\Rightarrow$). The arrow function is an exciting feature of ES6, loved by most JS enthusiasts.

The *for...of* iterator

JavaScript has *for* and *for...in* control statements to loop through a block of code a number of times and traverse through properties of an object, respectively. Note that *for...in* loops through the properties of an object. In ES6, a new looping structure *for...of* is introduced to loop through values of a collection. In addition to strings and arrays, we have Maps and Sets introduced in ES6.

*Snippet 1: Example of an existing *for...in* and new *for...of* looping statement*

```
1. let arr = ["mon", "tue", "wed"];
2. for (let i in arr) {
3.   console.log(i); //prints index
4. }
5. for (let i of arr) {
6.   console.log(i); //prints values
7. }
8. for (let [i,v] of arr.entries()) {
9.   console.log(i, ":", v); // prints index, values
10.}
```

Output:

```
1
2
mon
tue
wed
0: mon
1: tue
2: wed
```

In Lines 2-4, we have the looping structure, which displays only the index in case of arrays and property names in case of objects. In Lines 5-8, the new *for...of* statement assigns the value of the array to *i*. Lines 8-10 show another variation, where we have both the index and the value captured.

*Snippet 2: String object iterated using *for...of* loop*

```
1. var str = "ES6";
2. for (let i of str) {
3.   console.log(i);
4. }
```

Output:

```
E
S
6
```

